

ELECTIVE COURSE-6.5

URA753: ELECTRIC AND HYBRID VEHICLES				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: This course introduces the fundamental concepts, principles, analysis and design of electric and hybrid vehicles components. Students will also learn about the performance of electric and hybrid vehicles as compared to internal combustion engines.				
Syllabus Economic and Environmental Impact of Alternative Vehicle Options: Introduction and analysis of fuel usage; Technical and economical criteria; Environmental impact criteria; Correlation of alternate energy conversion. Electric Vehicles: History of electric vehicles; impact of modern drive-trains on energy supplies; Configurations of Electric Vehicles; Performance of Electric Vehicle Concept of Hybrid Electric Drive Trains; Traction Motor Characteristics; Tractive Effort and Transmission Requirement; Vehicle Performance; Tractive Effort in Normal Driving; Energy Consumption. Hybrid Electric Vehicles: Concept of Hybrid Electric Drive Trains; Architectures of Hybrid Electric Drive Trains; Series Hybrid Electric Drive Trains; Series Hybrid Electric Drive Trains; Parallel Hybrid Electric Drive Trains; Torque-Coupling and Speed-Coupling of Drive Trains. Electric Propulsion System and Sizing of EV components: Introduction to DC Motor Drives; Induction Motor Drives; Permanent Magnetic Brush-Less DC Motor Drives; Switched Reluctance Motor Drives; Control Strategies; Thermostat Control Strategy (Engine-On-Off); Power Rating Design of the Traction Motor; Power Rating Design of the Engine/Generator; Design of PPS; and few design examples. Battery Requirements for EVs and HEVs and Energy Storage: Energy requirements; cost; life of battery; temperature control; safety; specific requirements for micro-hybrids; mild-hybrids; full- hybrids; plugin-hybrids; Electrochemical Batteries; Electrochemical Reactions; Thermodynamic voltage; Specific energy; Specific power; Energy efficiency; Lithium-Based Batteries; Ultracapacitors based energy storage; Ultrahigh-Speed Flywheels; Hybridization of Energy Storages. Fundamentals of Regenerative Braking: Energy Consumption in Braking; Braking Power and Energy on Front and Rear Wheels; Brake System of EVs and HEVs; Antilock Brake System in Evs.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Case Studies: Case studies of few Electric and Hybrid vehicles.

Course Learning Outcomes (CLOs)

The students will be able to:

1. to analyze the performance of electric and hybrid.
2. analyze different powertrain topology used for electric vehicle application.
3. design of control unit for electric propulsion.

Text Books

1. Mehrdad E., Yimi G., Sebastian E. Gay, Ali Emadi, *Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design*, 2nd Edition, CRC Press (2009).
2. Gianfranco P., *Electric and Hybrid Vehicles: Power Source, Models, Sustainability, Infrastructure and the Market*, 1st Edition, Elsevier (2010).
3. Iqbal H., *Electric and Hybrid Vehicles: Design Fundamentals*, 2nd Edition, CRC Press (2011).
4. James Larminie, John Lowry, *Electric Vehicle Technology Explained*, 2nd Edition, Wiley (2012).

Reference Books

1. Sheldon S. Williamson, *Energy Management Strategies for Electric and Plug-in Hybrid Electric Vehicles*, 1st Edition, Springer (2013).
2. C.C. Chan and K.T. Chau, *Modern Electric Vehicle Technology*, 1st Edition, OXFORD University Press (2001).
3. Chris Mi, M. Abul Masrur, David Wenzhong Gao, *Hybrid Electric Vehicles Principles and Applications with Practical Perspectives*, 1st Edition, Wiley Publication (2011).

Evaluation Scheme:

S.No	Assessment	Weightage (%)
1	MST	35
2	EST	35
3	Sessionals	30

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